

17) Use truth tables to verify the absorption laws

a) $P \vee (P \wedge Q) \equiv P$

P	Q	$P \wedge Q$	$P \vee (P \wedge Q)$
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F

b) $P \wedge (P \vee Q) \equiv P$

P	Q	$P \vee Q$	$P \wedge (P \vee Q)$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	F

18) Determine whether $(\neg P \wedge (P \rightarrow Q)) \rightarrow \neg Q$ is a tautology.

Suppose for the sake of contradiction, it is not a tautology. $\therefore$ There exists an assignment of truth values of P, Q such that $(\neg P \wedge (P \rightarrow Q)) \rightarrow \neg Q$ is False. By the defn. of conditional, $(\neg P \wedge (P \rightarrow Q))$ is True and $\neg Q$ is False. By the defn. of negation, Q is True. By the defn. of AND, $\neg P$ is True and $P \rightarrow Q$ is True. Since Q is True, by the defn. of conditional, $(P \rightarrow Q)$ is True, which is consistent. By the defn. of negation, P is False.

$\therefore$ If P is False and Q is True, $(\neg P \wedge (P \rightarrow Q)) \rightarrow \neg Q$ is False.

So it's not a tautology.

19) Determine whether $(\neg Q \wedge (P \rightarrow Q)) \rightarrow \neg P$ is a tautology.

Suppose for the sake of contradiction, it is not a tautology. $\therefore$ There exists an assignment of truth values of P, Q such that $(\neg Q \wedge (P \rightarrow Q)) \rightarrow \neg P$ is False. By defn. of conditional, $\neg Q \wedge (P \rightarrow Q)$ is True but $\neg P$ is False. By defn. of negation, P is True. By defn. of AND, $\neg Q$ is True and $(P \rightarrow Q)$ is True. By defn. of negation, Q is False. Since P is True, Q is False, by the defn. of conditional, $(P \rightarrow Q)$ is False. This is a contradiction.

$\therefore$ It's a tautology